

Quantum Eraser and Information

April 9, 2026

1 Introduction

The quantum eraser experiment is a refined version of the double-slit experiment that explores the relationship between **information** and physical outcomes. It demonstrates that the presence or absence of **which-path information** determines whether interference patterns appear.

This raises a profound question:

Does information itself play a fundamental role in physical reality?

2 Basic Idea of the Experiment

In the standard double-slit experiment:

- If no which-path information is available, an interference pattern appears.
- If which-path information is known, the interference pattern disappears.

The quantum eraser experiment modifies this by:

- Marking paths with distinguishable information
- Then “erasing” that information

3 Which-Path Information

Which-path information refers to knowledge about which slit a particle passes through.

- **Available information** $\Rightarrow$ no interference
- **Unavailable information** $\Rightarrow$ interference

Importantly, it is the **availability in principle**, not actual observation, that matters.

4 Quantum Description

Suppose two paths are labeled:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle) \quad (1)$$

If path information is recorded in another system (environment or detector):

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle \otimes |A\rangle + |2\rangle \otimes |B\rangle) \quad (2)$$

where $|A\rangle$ and $|B\rangle$ are distinguishable states.

Tracing out the auxiliary system removes interference.

5 Quantum Erasure

If the auxiliary states are recombined such that:

$$|A\rangle, |B\rangle \rightarrow \text{indistinguishable} \quad (3)$$

then interference can be restored.

Key idea:

- Interference depends on distinguishability of information
- Erasing information restores coherence

6 Delayed Choice Aspect

In some versions of the experiment:

- The decision to erase or preserve information is made after detection

This suggests:

- No simple classical picture of cause and effect
- Correlations depend on the total experimental setup

However, it does not imply backward causation.

7 Implications for Physical Reality

7.1 Information as a Physical Quantity

The experiment suggests:

- Information is not abstract
- It has measurable physical consequences

7.2 No Role for Conscious Observation

Crucially:

- Interference disappears when information is recorded, even if no one observes it
- Consciousness is not required

7.3 Relational Nature of Reality

The results indicate:

- Physical outcomes depend on correlations between systems
- Reality may be relational rather than absolute

8 Connection to Decoherence

The quantum eraser highlights the role of decoherence:

- Environment stores which-path information
- Loss of coherence mimics classical behavior
- Erasure restores quantum features under controlled conditions

9 What the Experiment Does NOT Show

- No evidence of retrocausality
- No need for observer consciousness
- No violation of causality or relativity

10 Conclusion

The quantum eraser experiment reveals that:

- Interference depends on the availability of which-path information
- Information has a direct physical role in determining outcomes
- Quantum systems are fundamentally about correlations and coherence

In quantum mechanics, it is not simply particles that determine reality, but the structure of information and correlations between systems.

11 Multiple Choice Questions (MCQs)

Questions

1. What determines whether interference appears?
 - (a) Particle mass
 - (b) Which-path information availability
 - (c) Temperature
 - (d) Energy level
2. What happens when which-path information is available?
 - (a) Interference increases
 - (b) Interference disappears
 - (c) Nothing changes
 - (d) Energy increases
3. What does the quantum eraser do?
 - (a) Destroys particles
 - (b) Erases energy
 - (c) Removes distinguishability of paths
 - (d) Changes mass
4. Does consciousness play a role in the quantum eraser?
 - (a) Yes
 - (b) No
 - (c) Only sometimes
 - (d) Only in delayed choice
5. What happens when path information is erased?
 - (a) Interference is restored
 - (b) Particle disappears
 - (c) Energy is lost
 - (d) Time reverses
6. What does the delayed choice version suggest?
 - (a) Backward causation
 - (b) Dependence on total setup

- (c) Violation of relativity
 - (d) Faster-than-light signaling
7. What role does the environment play?
- (a) None
 - (b) Stores information leading to decoherence
 - (c) Increases energy
 - (d) Stops evolution
8. What is the deeper implication?
- (a) Only particles matter
 - (b) Information and correlations are fundamental
 - (c) Time is irrelevant
 - (d) Space disappears

Answers

- 1. b
- 2. b
- 3. c
- 4. b
- 5. a
- 6. b
- 7. b
- 8. b